

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12406051
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Sharma; Dinesh et al.

Password management of virtual machines for migration support

Abstract

Examples of the disclosure are directed to password management of virtual machines in a virtual machine environment. A password can be obtained that corresponds to a virtual machine. The password can be associated with a password identifier that is generated independent of a storage location of the files associated with the virtual machine. An operating system password vault can be utilized that stores the password and password identifier.

Inventors: Sharma; Dinesh (Bangalore, IN), Sahu; Amiya (Bangalore, IN), Garg; Himanshu (Bangalore, IN), Sheldon; Jeffrey W. (Los Altos, CA)

Applicant: VMware LLC (Palo Alto, CA)

Family ID: 94372088

Assignee: VMware LLC (Palo Alto, CA)

Appl. No.: 18/378158

Filed: October 10, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250036746 A1	Jan. 30, 2025

Foreign Application Priority Data

IN	202341049838	Jul. 24, 2023
----	--------------	---------------

Publication Classification

Int. Cl.: G06F21/45 (20130101); G06F9/455 (20180101)

U.S. Cl.:

Field of Classification Search

CPC: G06F (21/45); G06F (21/46); G06F (2009/45587); G06F (9/45558)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2010/0306773	12/2009	Lee	718/1	G06F 9/5077
2022/0391490	12/2021	Maresh	N/A	H04L 63/083
2023/0231833	12/2022	Cook	713/168	H04L 63/0272
2024/0405980	12/2023	Chang	N/A	H04L 9/3247

Primary Examiner: Jean; Frantz B

Attorney, Agent or Firm: King Intellectual Asset Management

Background/Summary

RELATED APPLICATIONS

(1) Benefit is claimed under 35 U.S.C. 119(a)-(d) to Foreign application No. 202341049838 filed in India entitled “PASSWORD MANAGEMENT OF VIRTUAL MACHINES FOR MIGRATION SUPPORT”, on Jul. 24, 2023, by VMware, Inc., which is herein incorporated in its entirety by reference for all purposes.

BACKGROUND

(2) Virtual machines are often utilized in a desktop environment so that a user, such as a developer, can utilize virtual machines for various purposes. For example, a developer might require different permutations of processor, memory, and operating system to run in an environment to test code that the developer is writing. As another example, a user might require a virtual machine running a different operating system other than a host operating system to perform work tasks. For example, a Mac user might require a Windows, Linux or Kubernetes environment to develop code that is being deployed to customers or perform other work-related tasks.

(3) A desktop environment can be provided in which a user can specify the configuration of a given virtual machine and save the configuration to the host operating system environment for later and subsequent use. The current or last working state of the virtual machine can also be saved for later use. However, security measures might be desired so that the state of the virtual machines is not accessible by other users of the host environment without appropriate credentials to do so.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily drawn to scale, with emphasis instead being placed upon clearly illustrating the principles of the disclosure. In the drawings, like reference numerals designate corresponding parts throughout the several views.

- (2) FIG. 1 illustrates an example networked environment for distributing and managing managed virtual machines and virtualized applications, according to various examples described herein.
- (3) FIG. 2 illustrates an example of packaging a managed virtual machine package and provisioning the managed virtual machine with virtualized applications, according to various examples described herein.
- (4) FIG. 3 illustrates a flowchart performed by components of the networked environment, according to various examples described herein.
- (5) FIG. 4 illustrates another flowchart performed by components of the networked environment, according to various examples described herein.

DETAILED DESCRIPTION

(6) The present disclosure relates to the security of virtual machines (VMs) in a desktop or development environment. As outlined above, personal devices are being used to access enterprise applications, data, and other resources with greater frequency. The use of these personal devices can be associated with productivity gains and cost savings. The concept “bring your own device” (BYOD) for access to enterprise resources faces a number of issues, including the provision for offline access. Whether a device is provided by the enterprise or the end user, devices are more often operated from home and other locations that are remote from an enterprise local area network. Application delivery can be problematic for BYOD scenarios. Files and registry keys remain on a device, file type associations are not reverted properly, services and resources shared with other applications are incorrectly removed, services are not properly unregistered, application data and log files not removed, or an uninstallation process fails without proper completion. Re-installing the same application has issues with these leftover artifacts. This can be unacceptable for enterprise and end users alike. However, the present disclosure provides a more effective solution that can provide both greater flexibility and isolation for virtualized application delivery in enterprise environments.

(7) Referring next to FIG. 1, shown in a host device **106**, which can represent a computing device or system in which examples of the disclosure can be implemented. Other examples or representations of a computing device can also be utilized to implement and understand examples of the disclosure. A host device **106** can be representative of one or more computing device. A host device **106** can be embodied as any computing device, processing circuit, or processor based device or system, including those in the form of a desktop computer, a laptop computer, a tablet computer, a personal digital assistant, a cellular telephone, a wearable computing device, or a set-top box, among other example computing devices and systems. Depending upon its primary purpose or function, for example, the host devices **106** can include various peripheral devices or components. The peripheral devices can include input or communications devices or modules, such as keyboards, keypads, touch pads, touch screens, microphones, cameras, wireless communications modules (e.g., infra-red, WI-FI, or BLUETOOTH®), buttons, switches, or sensors. The peripheral devices can also include a display, indicator lights, speakers, global positioning system (GPS) circuitry, accelerometers, gyroscopes, or other peripheral devices depending upon the primary purpose or function of the host devices **106**.

(8) A host device **106** can be equipped with various hardware resources and capabilities that can be made available to applications running on the host device **106** by an operating system. For example, a networking capability can allow applications to communicate with a network interface to access the Internet, intranets, extranets, wide area networks (WANs), local area networks (LANs), wired networks, wireless networks, cable networks, satellite networks, other suitable networks, or any combinations thereof. As one example, the host device **106** can be coupled to one or more public or private LANs or WANs and, in turn, to the Internet for communication of data among each other.

(9) The host device **106** can also include processor, mass storage, and memory resources that can be made available to applications. Accordingly, an application running on the host device **106** can

access the hardware resources of the host device **106** through calls made to an operating system that manages the hardware resources of the host device **106**. The hardware resources of the host device **106** are referred to herein as the host hardware **206**. The operating system can also provide other software application programming interfaces and services, such as a password vault **211**. A password vault **211** can comprise an operating system software service that permits applications and users of the host device **106** to store credentials, such as usernames, passwords, certificates, biometric credentials, and other information that can be protected by the operating system.

(10) The password vault **211** can require a user to authenticate himself or herself to the operating system before providing access to credentials stored therein. The password vault **211** can comprise a database or data store can store various credential entries. In the context of this disclosure, a credential entry **213** can comprise a password **215** and a password identifier **217**. A password **215** can comprise a user-generated or automatically generated password, passcode, passphrase, or other credential that can be utilized to secure or encrypt a virtual machine (VM). The VM can be secured or encrypted by a virtual machine environment.

(11) By utilizing the password vault **211** provided by the operating system of the host device **106**, the virtual machine environment **203** can provide password security without having to implement a user authentication feature or infrastructure within the virtual machine environment **203**.

(12) The host device **106** can execute a virtual machine environment **203**, which can also include a hypervisor **209** that acts as a software layer that interacts with the operating system of the host device **106**. The hypervisor **209** is an application or software stack that allows for creating and running virtual machines. Accordingly, a hypervisor **209** can be configured to provide guest systems with a virtual operating platform, including virtualized hardware devices or resources, and manage the execution of guest operating systems within a virtual machine execution space provided by the hypervisor **209**. In some instances, a hypervisor **209** may be configured to run directly on the hardware of the host device **106** in order to control and manage the hardware resources of the host device **106** provided to the virtual machines resident on the host device **106**. For example, the hypervisor **209** could be responsible for allocated time slices for the usage of a host GPU to virtual machines that request or require access to the resources of the GPU. In other instances, the hypervisor **209** can be implemented as an application executed by an operating system executed by the host device **106**, in which case the virtual machines may run as a thread, task, or process of the hypervisor **209** or operating system. Depending on the implementation, the hypervisor **209** could provide full virtualization for system virtual machines, or operating-system level virtualization for isolated processes that act as virtual machines.

(13) A VM **121** can represent software emulations of computer systems. Accordingly, a VM **121** can provide the functionality of a physical computer sufficient to allow for installation and execution of an entire operating system and any applications that are supported or executable by the operating system. As a result, a VM **121** can be used as a substitute for a physical machine to execute one or more processes. In full virtualization implementations, the VM **121** can be implemented as a system virtual machine that emulates a complete host. In operating-system virtualization implementations, the VM **121** can be implemented as an isolated user-space instance or process, which may be referred to as containers, zones, virtual environments, virtual kernels, or jails.

(14) Within the virtual machine environment **203**, a VM **121** can be associated with a VM configuration file **122**. The VM configuration file **122** can contain configuration information regarding a particular VM **121**. For example, the VM configuration file **122** can identify a processor and memory allocation, a type of GPU, an operating system that the VM **121** should execute, the mass storage resources of the VM **121**, and other configuration data associated with a VM **121**. In the context of this disclosure, the VM configuration file **122** can be stored on the host device **106** outside of the virtual machine environment **203**, such as in mass storage of the host device **106**. The VM configuration file **122** can also store a password identifier that is generated

based upon a password specified by a user for a password created by the user or automatically generated to protect or encrypt the VM **121**.

(15) A VM **121** can be protected or encrypted so that other users of the host device **106** are unable to access or execute the VM **121** without a password, passcode, or other type of credential. The virtual machine environment **203** can require the password to be provided at the time of accessing or execution of the VM **121**. In examples of the disclosure, the virtual machine environment **203** can utilize a password vault **211** of the operating system to store a password associated with a VM **121**. A credential entry **213** created by or on behalf of the virtual machine environment **203** can correspond to a particular VM **121**. The credential entry **213** can include a password **215** for a particular VM **121** as well as a password identifier **217**.

(16) A password identifier **217** can be generated based upon the password **215** and represent a unique identifier, or in some cases a globally unique identifier, that is created using the password **215** as an input. In some cases, the unique identifier can be based upon a timestamp taken when the password **215** was set for the VM **121**.

(17) The virtual machine environment **203** can store the password **215** and the password identifier **217** in a credential entry **213** within the password vault **211**. The password identifier **217** can be generated independently of a storage location of a VM **121** and its associated files on the host device **106**. For example, a state of the VM **121**, including the memory state, mass storage state, and other files needed to execute or resume the VM **121**, can be stored on the host device **106**. In some cases, the VM **121** can be associated with a directory or folder on the host device **106**.

(18) In some systems, when a password is stored in the password vault **211**, it is stored with an identifier the VM **121** to which the password corresponds is identified by its configuration file path or using a hash that is generated using the configuration file path. Accordingly, if the VM **121** files or a configuration file is moved or copied to a different location the identifier in the password vault **211** identifying the VM **121** does not work. A migration operation could comprise moving the VM to a different location, copying the VM **121**, exporting the VM **121**. The migration operation can include exportin, importing, or cloning the VM **121** which changes the VM configuration file **122** path or the path of any other files associated with a VM **121**.

(19) According to examples of the disclosure, the virtual machine environment **203** can generate a password identifier **217** that is not based upon the file path of the VM configuration file **122**, thereby breaking the binding between the password identifier **217** and the storage location of the VM configuration file **122** or the VM **121**. Accordingly, if a user or other system changes the password **215**, the virtual machine environment **203** can generate a new password identifier **217** that is stored along with the newly generated password **215** in the password vault **211**. Additionally, the password identifier **217** can be stored in the VM configuration file **122** so that the virtual machine environment **203**, when querying the password vault **211** for a password **215** corresponding to a particular VM **121**, can utilize the password identifier **217** stored in the VM configuration file **122** to locate the correct credential entry **213** in the password vault **211** to retrieve the correct password **215** for the VM **121**. In some examples, the password identifier **217** can be created based upon the password **215** at the time of encryption of a VM **121** within the virtual machine environment **203**.

(20) FIG. 2 shows a flowchart performed by components of the virtual machine environment **203**. Generally, the flowchart illustrates how the virtual machine environment **203** can generate a password identifier **217** and store the password identifier **217** in the password vault **211** of the operating system of the host device **106**. The virtual machine environment **203** can also store the password identifier **217** in a VM configuration file **122** corresponding to a VM **121**.

(21) First, at step **303**, the virtual machine environment **203** can execute a VM **121** within the virtual machine environment **203**. The VM **121** can be configured by a user with a selected amount of memory, virtual processing resources, virtual graphics processing resources, mass storage resources, a selected guest operating system of the VM **121**, and VM **121** properties. The

configuration of the VM **121** that is selected by the user can be saved into a VM configuration file **122** by the virtual machine environment **203**. The VM configuration file **122** can be saved into a folder or directory on the host device **106**.

(22) At step **306**, the virtual machine environment **203** can obtain a password **215** for the VM **121**. The password **215** can be specified by the user or generated by the virtual machine environment **203** or a password manager. The password **215** can be used to encrypt or otherwise secure the VM **121** so that users without the password **215** are unable to access or execute the VM **121** within the virtual machine environment **203**.

(23) At step **309**, the virtual machine environment **203** can generate a password identifier **217** corresponding to the password **215** obtained at step **306**. The password identifier **217** can be a hash of the password **215** or a globally unique identifier that is generated using the password **215** as an input. The password identifier **217** can be utilized as an encryption key or a seed to generate a key with which the VM **121** can be encrypted or otherwise secured.

(24) At step **312**, the virtual machine environment **203** can store the password identifier **217** in a VM configuration file **122** corresponding to the VM **121**. By storing the password identifier **217** into the VM configuration file **122**, the virtual machine environment **203** can query the password vault **211** for the correct password **215** by searching the password vault **211** with the password identifier **217**. In this sense, the password identifier **217** can identify the VM **121** in the password vault **211** as well as the password **215**.

(25) At step **315**, the virtual machine environment **203** can store the password **215** and the password identifier **217** in the password vault **211** in a credential entry **213**. The credential entry **213** can represent an entry in the password vault **211** that is associated with the VM **121**. Using the password identifier **217**, again, the virtual machine environment **203** can query the password vault **211** for the correct password **215** by searching the password vault **211** with the password identifier **217**. In this sense, the password identifier **217** can identify the VM **121** in the password vault **211** as well as the password **215**. Thereafter, the process proceeds to completion.

(26) FIG. 3 shows a flowchart performed by components of the virtual machine environment **203**. Generally, the flowchart illustrates how the virtual machine environment **203** obtain a password identifier **217** from the password vault **211** of the operating system of the host device **106**. The virtual machine environment **203** can utilize the password identifier **217** to obtain the password **215** for a VM **121** so that the virtual machine environment **203** can execute or access the VM **121**.

(27) First, at step **403**, the virtual machine environment **203** can obtain a request to execute a VM **121** within the virtual machine environment **203**. The request can be generated in response to a user attempting to access or execute the VM **121** in a user interface provided by the virtual machine environment **203** on the host device **106**. The VM **121** can allow the user to utilize a VM **121** with a selected amount of memory, virtual processing resources, virtual graphics processing resources, mass storage resources, a selected guest operating system of the VM **121**. The configuration of the VM **121** that is selected by the user can be retrieved from a VM configuration file **122** by the virtual machine environment **203**. The VM configuration file **122** can indicate a password identifier **217** and/or whether the VM **121** is encrypted or secured by a password **215**.

(28) At step **405**, the virtual machine environment **203** can identify a password identifier **217** in the VM configuration file **122**. The password identifier **217** can be utilized by the virtual machine environment **203** to locate the password **215** for the VM **121** in the password vault **211**.

(29) At step **407**, the virtual machine environment **203** can submit a request to the password vault **211** for a credential entry **213** in which the password identifier **217** is present. In other words, the virtual machine environment **203** can search the password vault **211** for the credential entry **213** containing the password **215** that corresponds to the password identifier **217** and to the VM **121**. By utilizing the password vault **211** provided by the operating system, the virtual machine environment **203** need not implement its own password security or user authentication scheme, instead relying on the operating system for password security and user authentication. Examples of

a password vault **211** provided by an operating system include the Windows® Credential Manager and MacOS® Keychain. In some implementations, a third party password vault **211** can be utilized that is not integrated into the operating system, such as a third party password or credential manager that runs as an application on the host device **106**. The password vault **211** can provide the credential entry **213** containing the password **215** corresponding to the password identifier **217** to the virtual machine environment **203**.

(30) At step **409**, the virtual machine environment **203** can access the VM **121** using the password **215** retrieved from the password vault **211**. In one example, the virtual machine environment **203** can decrypt the VM **121** using the password **215** or a key derived from the password **215**. In another example, the virtual machine environment **203** can simply require the password **215** in order to access the VM **121**. At step **411**, the virtual machine environment **203** can execute or otherwise access the VM **121**. Thereafter, the process can proceed to completion.

(31) FIG. **4** shows a flowchart performed by components of the virtual machine environment **203**. Generally, the flowchart illustrates how the virtual machine environment **203** can update a password identifier **217** in the password vault **211** of the operating system of the host device **106** in response to a change of a password **215** corresponding to a VM **121**.

(32) First, at step **503**, the virtual machine environment **203** can obtain a password change for a password **215** corresponding to a VM **121** in the virtual machine environment **203**. The password change can comprise a new password that is obtained from a user or automatically generated by the virtual machine environment **203** or another application.

(33) At step **506**, the virtual machine environment **203** can generate a new password identifier **217** corresponding to the password **215** obtained in step **503**. The password identifier **217** can be a hash of the password **215** or a globally unique identifier that is generated using the password **215** as an input. The password identifier **217** can be utilized as an encryption key or a seed to generate a key with which the VM **121** can be encrypted or otherwise secured.

(34) At step **509**, the virtual machine environment **203** can store or update the password identifier **217** in a VM configuration file **122** corresponding to the VM **121**. By storing the password identifier **217** into the VM configuration file **122**, the virtual machine environment **203** can query the password vault **211** for the correct password **215** by searching the password vault **211** with the password identifier **217**. In this sense, the password identifier **217** can identify the VM **121** in the password vault **211** as well as the password **215**.

(35) At step **512**, the virtual machine environment **203** can store the password **215** and the password identifier **217** in the password vault **211** in a credential entry **213**. The credential entry **213** can represent an entry in the password vault **211** that is associated with the VM **121**. Using the password identifier **217**, again, the virtual machine environment **203** can query the password vault **211** for the correct password **215** by searching the password vault **211** with the password identifier **217**. In this sense, the password identifier **217** can identify the VM **121** in the password vault **211** as well as the password **215**. Thereafter, the process proceeds to completion.

(36) The flowchart(s) and sequence diagram(s) show examples of the functions and operation of the components described herein. The components described herein can be embodied in hardware, software, or a combination of hardware and software. If embodied in software, each element can represent a module or group of code that includes program instructions to implement the specified logical function(s). The program instructions can be embodied in the form of, for example, source code that includes human-readable statements written in a programming language or machine code that includes machine instructions recognizable by a suitable execution system, such as a processor in a computer system or other system. If embodied in hardware, each element can represent a circuit or several interconnected circuits that implement the specified logical function(s).

(37) The computing environment **103** can include at least one processing circuit. Such a processing circuit can include, for example, one or more processors and one or more storage or memory devices that are coupled to a local interface. The local interface can include, for example, a data bus

with an accompanying address/control bus or any other suitable bus structure. Similarly, the host devices **106** can each include at least one processing circuit. Such a processing circuit can include, for example, one or more processors and one or more storage or memory devices that are coupled to a local interface.

(38) The storage or memory devices can store data or components that are executable by the processors of the processing circuit. For example, the management service **120** and/or other components can be stored in one or more storage devices and be executable by one or more processors in the networked environment **100**. Similarly, the agents, services, applications and/or other components described herein can be stored in one or more storage devices and be executable by one or more processors in the host device **106**.

(39) The management service **120** and/or other components described herein can be embodied in the form of hardware, software components that are executable by hardware, or as a combination of software and hardware. If embodied as hardware, the components described herein can be implemented as a circuit or state machine that employs any suitable hardware technology. The hardware technology can include, for example, one or more microprocessors, discrete logic circuits having logic gates for implementing various logic functions upon an application of one or more data signals, application specific integrated circuits (ASICs) having appropriate logic gates, programmable logic devices (e.g., field-programmable gate array (FPGAs), and complex programmable logic devices (CPLDs)).

(40) Also, one or more of the components described herein that include software or program instructions can be embodied in any non-transitory computer-readable medium for use by or in connection with an instruction execution system such as, a processor in a computer system or other system. The computer-readable medium can contain, store, and/or maintain the software or program instructions for use by or in connection with the instruction execution system.

(41) A computer-readable medium can include a physical media, such as, magnetic, optical, semiconductor, and/or other suitable media. Examples of a suitable computer-readable media include, but are not limited to, solid-state drives, magnetic drives, or flash memory. Further, any logic or component described herein can be implemented and structured in a variety of ways. For example, one or more components described can be implemented as modules or components of a single application. Further, one or more components described herein can be executed in one computing device or by using multiple computing devices.

(42) Further, any logic or applications described herein, including the management service **120** and/or other components can be implemented and structured in a variety of ways. For example, one or more applications described can be implemented as modules or components of a single application. Further, one or more applications described herein can be executed in shared or separate computing devices or a combination thereof. For example, a plurality of the applications described herein can execute in the same computing device, or in multiple computing devices. Additionally, terms such as “application,” “service,” “system,” “engine,” “module,” and so on can be used interchangeably and are not intended to be limiting.

(43) The above-described examples of the present disclosure are merely possible examples of implementations set forth for a clear understanding of the principles of the disclosure. Many variations and modifications can be made without departing substantially from the spirit and principles of the disclosure. All such modifications and variations are intended to be included herein within the scope of this disclosure and protected by the following claims.

(44) Certain embodiments as described above involve a hardware abstraction layer on top of a host computer. The hardware abstraction layer allows multiple containers to share the hardware resource. These containers, isolated from each other, have at least a user application running therein. The hardware abstraction layer thus provides benefits of resource isolation and allocation among the containers. In the foregoing embodiments, virtual machines are used as an example for the containers and hypervisors as an example for the hardware abstraction layer. As described

above, each virtual machine includes a guest operation system in which at least one application runs. It should be noted that these embodiments may also apply to other examples of containers, such as containers not including a guest operating system, referred to herein as “OS-less containers” (see, e.g., www.docker.com). OS-less containers implement operating system-level virtualization, wherein an abstraction layer is provided on top of the kernel of an operating system on a host computer. The abstraction layer supports multiple OS-less containers each including an application and its dependencies. Each OS-less container runs as an isolated process in userspace on the host operating system and shares the kernel with other containers. The OS-less container relies on the kernel's functionality to make use of resource isolation (CPU, memory, block I/O, network, etc.) and separate namespaces and to completely isolate the application's view of the operating environments. By using OS-less containers, resources can be isolated, services restricted, and processes provisioned to have a private view of the operating system with their own process ID space, file system structure, and network interfaces. Multiple containers can share the same kernel, but each container can be constrained to only use a defined amount of resources such as CPU, memory and I/O.

(45) The embodiments described herein can employ various computer-implemented operations involving data stored in computer systems. For example, these operations can require physical manipulation of physical quantities-usually, though not necessarily, these quantities take the form of electrical or magnetic signals, where they (or representations of them) are capable of being stored, transferred, combined, compared, or otherwise manipulated. Such manipulations are often referred to in terms such as producing, identifying, determining, comparing, etc. Any operations described herein that form part of one or more embodiments can be useful machine operations.

(46) Further, one or more embodiments can relate to a device or an apparatus for performing the foregoing operations. The apparatus can be specially constructed for specific required purposes, or it can be a general purpose computer system selectively activated or configured by program code stored in the computer system. In particular, various general purpose machines may be used with computer programs written in accordance with the teachings herein, or it may be more convenient to construct a more specialized apparatus to perform the required operations. The various embodiments described herein can be practiced with other computer system configurations including handheld devices, microprocessor systems, microprocessor-based or programmable consumer electronics, minicomputers, mainframe computers, and the like.

(47) Yet further, one or more embodiments can be implemented as one or more computer programs or as one or more computer program modules embodied in one or more non-transitory computer readable storage media. The term non-transitory computer readable storage medium refers to any data storage device that can store data which can thereafter be input to a computer system. The non-transitory computer readable media may be based on any existing or subsequently developed technology for embodying computer programs in a manner that enables them to be read by a computer system. Examples of non-transitory computer readable media include a hard drive, network attached storage (NAS), read-only memory, random-access memory, flash-based nonvolatile memory (e.g., a flash memory card or a solid state disk), a CD (Compact Disc) (e.g., CD-ROM, CD-R, CD-RW, etc.), a DVD (Digital Versatile Disc), a magnetic tape, and other optical and non-optical data storage devices. The non-transitory computer readable media can also be distributed over a network coupled computer system so that the computer readable code is stored and executed in a distributed fashion.

(48) In addition, while described virtualization methods have generally assumed that virtual machines present interfaces consistent with a particular hardware system, persons of ordinary skill in the art will recognize that the methods described can be used in conjunction with virtualizations that do not correspond directly to any particular hardware system. Virtualization systems in accordance with the various embodiments, implemented as hosted embodiments, non-hosted embodiments or as embodiments that tend to blur distinctions between the two, are all envisioned.

Furthermore, certain virtualization operations can be wholly or partially implemented in hardware.

(49) Many variations, modifications, additions, and improvements are possible, regardless the degree of virtualization. The virtualization software can therefore include components of a host, console, or guest operating system that performs virtualization functions. Plural instances can be provided for components, operations, or structures described herein as a single instance. Finally, boundaries between various components, operations, and data stores are somewhat arbitrary, and particular operations are illustrated in the context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within the scope of the invention(s). In general, structures and functionality presented as separate components in exemplary configurations can be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component can be implemented as separate components.

(50) As used in the description herein and throughout the claims that follow, “a,” “an,” and “the” includes plural references unless the context clearly dictates otherwise. Also, as used in the description herein and throughout the claims that follow, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

(51) The above description illustrates various embodiments along with examples of how aspects of particular embodiments may be implemented. These examples and embodiments should not be deemed to be the only embodiments, and are presented to illustrate the flexibility and advantages of particular embodiments as defined by the following claims. Other arrangements, embodiments, implementations and equivalents can be employed without departing from the scope hereof as defined by the claims.

Claims

1. A method implemented by instructions executed by at least one computing device, the method comprising: executing, within a host device comprising a host operating system, a virtual machine comprising a guest operating system and the host device executes a virtual machine environment in which the virtual machine executes; obtaining, within the virtual machine environment on the host device, a password for securing the virtual machine; generating, within the virtual machine environment, a virtual machine configuration file for the virtual machine; generating, within the virtual machine environment, a password identifier corresponding to the password and the virtual machine; storing, within the virtual machine environment, the password identifier in the virtual machine configuration file; and storing, by the virtual machine environment, the password and the password identifier in a password vault provided by the host operating system.
2. The method of claim 1, further comprising: obtaining, within the virtual machine environment, a request to execute the virtual machine; identifying, within the virtual machine environment, the password identifier corresponding to the virtual machine in the configuration file corresponding to the virtual machine; identifying, in the password vault provided by the host operating system, the password identifier identified in the virtual machine configuration file; retrieving, from the password vault provided by the host operating system, the password corresponding to the password identifier; accessing, within the virtual machine environment, the virtual machine using the password retrieved from the password vault provided by the host operating system.
3. The method of claim 1, further comprising: receiving, within the virtual machine environment, a request to modify the password corresponding to the virtual machine to a new password; generating, within the virtual machine environment, a new password identifier corresponding to the new password; and updating, within the virtual machine environment, the virtual machine configuration file with the new password identifier.
4. The method of claim 1, wherein the password identifier is generated independent of and without a storage location of the virtual machine configuration file as an input.
5. The method of claim 1, wherein the password identifier comprises a globally unique identifier.

6. The method of claim 1, further comprising encrypting, by the virtual machine environment, the virtual machine based upon the password.
7. The method of claim 1, further comprising: generating, by the virtual machine environment, a copy of the virtual machine, wherein the copy of the virtual machine is secured using the password.
8. A non-transitory computer-readable medium embodying instructions executable by at least one computing device wherein the instructions, when executed, cause the at least one computing device to at least: execute, within a host device comprising a host operating system, a virtual machine comprising a guest operating system and the host device executes a virtual machine environment in which the virtual machine executes; obtain a password for securing the virtual machine; generate a virtual machine configuration file for the virtual machine; generate a password identifier corresponding to the password and the virtual machine; store the password identifier in the virtual machine configuration file; and store the password and the password identifier in a password vault provided by the host operating system.
9. The non-transitory computer-readable medium of claim 8, wherein when executed, the instructions further cause the at least one computing device to at least: obtaining, within the virtual machine environment, a request to execute the virtual machine; identifying, within the virtual machine environment, the password identifier corresponding to the virtual machine in the configuration file corresponding to the virtual machine; identifying, in the password vault provided by the host operating system, the password identifier identified in the virtual machine configuration file; retrieving, from the password vault provided by the host operating system, the password corresponding to the password identifier; accessing, within the virtual machine environment, the virtual machine using the password retrieved from the password vault provided by the host operating system.
10. The non-transitory computer-readable medium of claim 8, wherein when executed, the instruction further cause the at least one computing device to at least: receive a request to modify the password corresponding to the virtual machine to a new password; generate a new password identifier corresponding to the new password; and update the virtual machine configuration file with the new password identifier.
11. The non-transitory computer-readable medium of claim 8, wherein the password identifier is generated independent of and without a storage location of the virtual machine configuration file as an input.
12. The non-transitory computer-readable medium of claim 8, wherein the password identifier comprises a globally unique identifier.
13. The method according to claim 8, wherein when executed, the instructions further cause the at least one computing device to at least encrypt the virtual machine based upon the password.
14. The method according to claim 8, wherein when executed, the instructions further cause the at least one computing device to at least generate a copy of the virtual machine, wherein the copy of the virtual machine is secured using the password.
15. A system, comprising: at least one computing device; and instructions accessible by the at least one computing device, wherein when executed the instructions cause the at least one computing device to at least: execute, within a host device comprising a host operating system, a virtual machine comprising a guest operating system and the host device executes a virtual machine environment in which the virtual machine executes; obtain a password for securing the virtual machine; generate a virtual machine configuration file for the virtual machine; generate a password identifier corresponding to the password and the virtual machine; store the password identifier in the virtual machine configuration file; and store the password and the password identifier in a password vault provided by the host operating system.
16. The system of claim 15, wherein when executed, the instructions further cause the at least one computing device to at least: obtaining, within the virtual machine environment, a request to execute the virtual machine; identifying, within the virtual machine environment, the password

identifier corresponding to the virtual machine in the configuration file corresponding to the virtual machine; identifying, in the password vault provided by the host operating system, the password identifier identified in the virtual machine configuration file; retrieving, from the password vault provided by the host operating system, the password corresponding to the password identifier; and accessing, within the virtual machine environment, the virtual machine using the password retrieved from the password vault provided by the host operating system.

17. The system of claim 15, wherein when executed the instructions further cause the at least one computing device to at least: receive a request to modify the password corresponding to the virtual machine to a new password; generate a new password identifier corresponding to the new password; and update the virtual machine configuration file with the new password identifier.

18. The system of claim 15, wherein the password identifier is generated independent of and without a storage location of the virtual machine configuration file as an input.

19. The system of claim 18, wherein when executed, the instructions further cause the at least one computing device to at least encrypt the virtual machine based upon the password.

20. The system of claim 19, wherein when executed, the instructions further cause the at least one computing device to at least generate a copy of the virtual machine, wherein the copy of the virtual machine is secured using the password.
